

Team LONG PULL — 2026-07-08 (eve). RETRACT the quark-mass floor (premature). The flavor sector is ONE object — the generation positions. Force them → masses + mixing close together. Use BST's OWN values, not λ QCD. date + SOD.

Casey's steer + the retractions (absorb honestly — this reverses two of our own conclusions)

- CASEY: use BST ideas/values, NOT λ QCD (they overlap sometimes, not always; the quark masses may yet derive). → The quark-mass “floor” is RETRACTED as premature. Two λ QCD imports made it look like a floor: (a) the target {1,20,900} is MS-bar, scheme-dependent — comparing BST's geometric spectrum to it directly is apples-to-oranges; (b) the “QCD-dominated” mechanism was external. The masses are OPEN, not floored. (Keeper's floor-note is SHELVED.)
- LYRA's one-gate correction: mixing and masses are the SAME object — one set of generation positions (spacing → masses, overlaps → mixing). Cannot bank mixing “forward” while masses “floor” — one matrix, one gate. Both close or both stay open. → Retract “mixing forward, quarks floor.”
- ELIE's lepton test: the upgrades don't cleanly justify (τ/e FAILS — wrong sign, $8\times$ too big; μ/e weak, π^2 -ambiguous). Leptons aren't a clean upgrade either. The mass sector is uniformly gated on the positions.

The unifying picture (honest, revised)

The flavor sector = ONE object: the three generation positions in the ball. - The mixing MACHINERY is clean (Lyra's hyperbolic engine: positions {0, 1, n_C/rank } → all angles, few-to-10%). - The masses come from the SAME positions (the spacing — NOT $d(\nu)$, which failed robustly). - THE GATE = force the positions. Force → both masses and mixing close. Neither floored; both gated on (the positions) + (the correct spacing formula) + (BST's own scheme).

PRIMARY — force the generation positions (the ONE gate) · LYRA + GRACE

Force ONE of the candidate structures for the positions: - (a) Lyra's engine: $r_2 = 1/\text{rank}^2$, gaps = n_C/rank per step → {0, 1, n_C/rank }. - (b) The cube-root structure: 3 generations at 120° apart (cube roots — because there are exactly 3, the same 3 as the colors). Forced positions → BOTH the mixing (Lyra's overlaps: V_{us}, V_{cb}, V_{ub}) AND the masses (the spacing) close. This is the one move that unlocks the whole flavor sector.

GATE-INDEPENDENT FORWARD LEAD — CP violation forced by $N_c=3$ · LYRA (the exciting one)

The one piece that does NOT wait on the positions: if the 3 generations sit at cube-root/ 120° directions *because* $N_c=3$, they cannot be collinear → CP violation is guaranteed to EXIST, and its handedness = the triangle's orientation. This ties CP + the matter/antimatter asymmetry to the bare fact $N_c=3$ (the same winding/handedness thread from the CMB cosmology). Structural candidate: show the cube-root/ 120° structure is FORCED (the 3 generations ARE the 3 colors at cube-root directions) → “CP exists because $N_c=3$ ” banks as structure. Forward, gate-independent — push it.

BST-NATIVE MASSES (Casey's steer) · GRACE

Do NOT match the geometric spectrum to λ QCD MS-bar {1,20,900}. Instead: (a) what does BST's OWN geometry give for the quark spectrum from the forced positions? (b) what is BST's OWN scheme/conversion to the observable masses (NOT λ QCD running)? The apparent “over-determination near 20 and 900” may be the scheme conversion sitting on top of a clean BST-native spectrum. Derive BST-native spectrum + BST-native conversion — that's the un-tried path (we only ever tried to hit the scheme-dependent MS-bar values).

Per-CI

- LYRA: force the positions ($r_2=1/\text{rank}^2$ or the cube-root structure) — the one gate; and the CP-from- $N_c=3$ structural lead (gate-independent, forward). Both are yours and both are strong.

- GRACE: the positions (with Lyra) + the BST-native mass spectrum/conversion (Casey's steer — NOT λ QCD); then α root-lock #2 (does the count 137 = the $S^2 \times S^1$ coupling curvature = Wyler — confirms the trunk).
- ELIE: hold as diagonalizer (the four matrices close the instant the positions land); the μ/e QED calc only if cycles (weak, π^2 -ambiguous — low priority).
- KEEPER: adjudicate the forced positions + the CP-candidate + the BST-native masses; the floor is RETRACTED (shelved, not a result); track the count with Cal; over-sell #7.
- CAL: finalize the reconciled count (5 firm + α ; $\sin^2 \theta_{12}$ call).

Disciplines armed

Use BST's own values, not λ QCD (Casey) — the MS-bar mass target is scheme-suspect; derive BST-native. Mixing + masses = ONE gate (Lyra) — don't bank one ahead of the other. Force the positions (shown, not assumed — $r_2=1/\text{rank}^2$ or cube-root, from the geometry). CP-from- $N_c=3$ = candidate (show the cube-root forced; beautiful → hardest look). The floor is retracted — masses are OPEN, and “the quark masses will derive” is the working posture (Casey). Over-sell #7 watch. Count as Cal reconciles.

What could bank (forward)

- The positions forced → masses + mixing close TOGETHER (the big one — most of the flavor sector at once).
- CP-exists-from- $N_c=3$ → structural bank (gate-independent).
- α root-lock #2 → confirms the trunk.

Reserved for Casey

Mass-frame ratification · the paper (gated on the reconciled count).

Long-pull target

Force the generation positions (the one gate closing masses + mixing together). Land the CP-from- $N_c=3$ structural lead (gate-independent). Derive the quark masses in BST's OWN scheme (not λ QCD MS-bar). α root-lock #2 confirms the trunk. The floor is retracted; the masses are open and being worked forward, per Casey.

— Keeper, 2026-07-08 eve. RETRACT the quark-mass floor (premature — Casey: use BST values not λ QCD, the MS-bar target imports λ QCD; + Lyra's one-gate: masses+mixing are one object). The flavor sector = the generation positions; force them → both close. PRIMARY: force the positions ($r_2=1/\text{rank}^2$ or cube-root). GATE-INDEPENDENT LEAD: CP violation forced by $N_c=3$ (cube-root generations can't be collinear → CP guaranteed, handedness = triangle orientation; ties CP+matter/antimatter to $N_c=3$). BST-native masses (not λ QCD). α root-lock #2. Over-sell #7. Masses OPEN, working posture “they will derive.”